Problem Set 1.

1. (Setting up sample spaces) Assume two standard six-sided dice are rolled one after the other.

   i Restrict attention to the first roll, and list all possible outcomes of this random experiment.

   ii Now consider both rolls, and list the outcomes of the most general sample space you can define.

2. (Set theory) Consider two arbitrary events A, $B \in S$ and describe the following events using set operations.

   i Both events occur.

   ii At least one event occurs.

   iii Neither event occurs.

   iv Only event A but not event B occurs, i.e. $\{s \in S : s \in A \text{ and } s \notin B\}$.
   Note: this set operation is called *difference* and is denoted $A - B$ (minus) or $A \backslash B$ (backslash).

   v Exactly one event occurs.
   Note: this set operation is called *symmetric difference* and is denoted by $A \triangle B$. In logic it is also called *exclusive OR* (XOR).

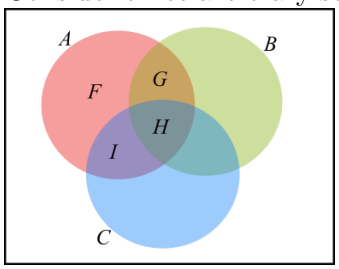
3. (Blitzstein: §1, Q42-43) For arbitrary events A, B use the probability axioms to show

   (a) $P(A - B) = P(A) - P(A \cap B)$.

   (b) $P(A \triangle B) = P(A) + P(B) - 2P(A \cap B)$.

   The definition of the difference $(-)$ and symmetric difference $(\triangle)$ set operations are given in problem 2.iv and 2.v.

4. Consider three arbitrary sets A, B, C represented by colored circles in the Venn diagram below.



   (a) Describe the strictly pink area F using set operations on A, B, C.

   (b) Describe the strictly brown area G using set operations on A, B, C.

(c) Show that $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$, i.e. prove the 3-set version of the inclusion-exclusion principle.

5. (Inclusion-exclusion principle; SOA Exam P, May 2003) A survey of a group's viewing habits over the last year revealed the following information:

   - 28% watched gymnastics (G)
   - 29% watched baseball (B)
   - 19% watched soccer (S)
   - 14% watched gymnastics and baseball ($G \cap B$)
   - 12% watched baseball and soccer ($B \cap S$)
   - 10% watched gymnastics and soccer ($G \cap S$)
   - 8% watched all three sports. ($G \cap B \cap S$)

   Calculate the percentage of the group that:

   i Watched only gymnastics last year.

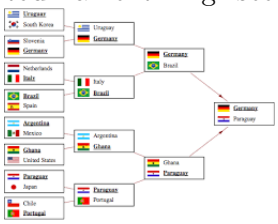
   ii Watched none of the three sports last year.

6. (Blitzstein: §1, Q4) A *round-robin tournament* is being held with n tennis players; this means that every player will play against every other player exactly once.

   (a) How many games are played in total?

   (b) How many possible outcomes are there for the tournament (the outcomes list who won and who lost for each game).

7. (Blitzstein: §1, Q5) A *knock-out* tournament is being held with 2^n players. This means that for each round, the winners move on to the next round and the losers are eliminated, until only one person remains. For example, if initially there are $2^4 = 16$ players, then there are 8 games in first round, then the 8 winners move on to round 2, then the 4 winners move on to round 3, then the 2 winners move in to round 4, the winner of which is declared the winner of the tournament. E.g. see below:



   (a) How many rounds are there for general 2^n players?

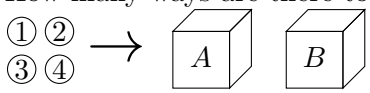
   (b) Count how many games are played in total, by adding up the number of games in each round (Hint: it is a geometric series).

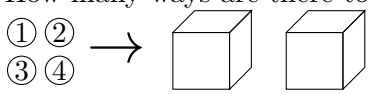
   (c) Count how many games are played in total, by considering the number of players that need to be eliminated.

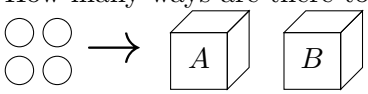
   (d) Consider a knock-out tournament with $n \geq 2$ where the two best players in the world participate. Assume these two players always win other players. If they are randomly assigned in the tournament table, what are the chances they will meet in the final (but not before)?

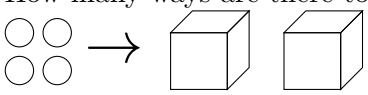
8. (Blitzstein: §1, Q8) How many ways are there to split 12 people into 3 teams, where:
   (a) One team has 2 people and the other two have 5 people each.
   (b) Every team has 4 people.

9. (Evans 1.4.2) Suppose we roll 10 fair six-sided dice.
   (a) What is the probability there are *exactly* two 6's showing?
   (b) What is the probability there are *at least* two 6's showing?

10. How many ways are there to put 4 *distinguishable* balls into 2 *distinguishable* boxes:


11. How many ways are there to put 4 *distinguishable* balls into 2 *indistinguishable* boxes:


12. How many ways are there to put 4 *indistinguishable* balls into 2 *distinguishable* boxes:


13. How many ways are there to put 4 *indistinguishable* balls into 2 *indistinguishable* boxes:


14. Consider a class consisting of 15 girls and 10 boys.
   (a) Assume the teacher wants to form an *unordered* pair consisting of 1 girl and 1 boy. How many ways are there to form such a pair?
   (b) Assume the teacher wants to form an *unordered* group of 3, consisting of 2 girls and 1 boy. How many ways are there to form such a group?
   (c) Assume the teacher now wants to form an *ordered* group of 3, consisting of 2 girls and 1 boy (e.g., think of each student having a *different* task, so their order, i.e. who does what, matters). How many ways are there to form such a group?

15. (Quiz 1; Fall 2022) Your baseball team is playing a best-of-five series, i.e a sequence of games in which the first team to win 3 games in total wins the entire series. Think of the whole series as a random experiment and represent its outcomes as sequences of wins or losses (denoted by W/L) for your team. E.g., the outcome (W, L, W, W) means that your team won games 1,3 and 4 (and lost game 2), at which point your team won the series. *Note that there are no more games played after a team wins the series.* Define the events:
   - $A =$ { your team won three games *in a row* }
   - $B =$ { your team won the series }
   - $C =$ { the opposing team won *at least* one game }

Answer the following questions:

(a) (2 points) List all the outcomes in event A.

(b) (3 points) What is the relationship of events A and B: are they disjoint, complementary, or is one a subset of the other? Justify your answer.

(c) (3 points) List all the outcomes in event $(B \cap C)$.

(d) (4 points) List all the outcomes in the sample space (S).
(Hint: there are 20 distinct outcomes)

For the remaining questions, we assign probabilities to outcomes according to the following scheme: Assume both teams have the same chances of winning each game, irrespective of other games. This implies that the probability of an outcome is equal to $(1/2)^\ell$, where ℓ is the *series length*. E.g., $P((W, L, W, W)) = (1/2)^4$, because there are 4 games played in the series (W, L, W, W). This defines a probability function that assigns to any event E the sum of the probabilities of all outcomes in E:

$$P(E) = \sum_{s_i \in E} P(\{s_i\})$$

(e) (4 points) Find the value of the probability $P(A)$.

(f) (4 points) Verify that $P(S) = 1$, i.e. that this is a valid probability function.

16. (Quiz 1, Summer 2025) Consider a probability model with arbitrary events A, B and probability function $\mathbb{P}(\cdot)$. Find $\mathbb{P}(A)$ if you know that $\mathbb{P}(A \cup B) = 0.6$ and $P(A \cup B^c) = 0.8$.

17. (Quiz 1, Fall 2023) Consider three events A, B and C. Let E be the event that two and only two of the three events occur. Visualize E using a Venn diagram and write down an expression of E in terms of A, B and C and appropriate set operations.

18. (Quiz 1, Fall 2023) Suppose a fair die is thrown twice and assume that each of the 36 outcomes occurs with the same probability.

    (a) Find the probability that the sum of the two numbers is at least $(\geq)$ 7 and that the two numbers are at least $(\geq)$ 2 apart. For example, the outcome (5,4) does not satisfy the requirement even though $5 + 4 > 7$.

    (b) Write down an event A such that: (i) the probability of A is $\frac{1}{3}$ and (ii) if $B = \{$ first number is $3\}$ then the probability of $A \cup B$ is $\frac{1}{2}$.

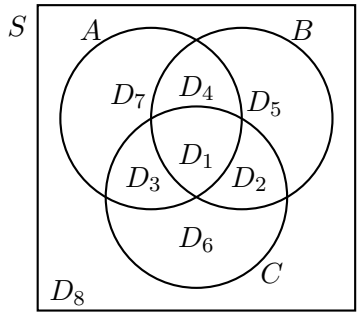
19. (Quiz 1, Summer 2024) Consider arbitrary events A, B, and a set function $\mathbb{P}(\cdot)$ such that $\mathbb{P}(A) = 0.3$, $\mathbb{P}(B) = 0.4$, $\mathbb{P}(A \cup B) = 0.8$. Is the function $\mathbb{P}(\cdot)$ a valid probability measure? Prove or disprove, based only on the axioms of probability and their derived laws.

20. (Quiz 1, Summer 2024) Consider a probability model with events A, B, C, and a probability function $\mathbb{P}(\cdot)$ such that:

$$\mathbb{P}(A) = 0.3,\ \mathbb{P}(B) = 0.4,\ \mathbb{P}(C) = 0.5$$
$$\mathbb{P}(A \cup B) = 0.55,\ \mathbb{P}(A \cup C) = \mathbb{P}(B \cup C) = 0.65$$
$$\mathbb{P}(A \cap B \cap C) = 0.1$$

Find $\mathbb{P}((A \cup B^c) \cap C)$.

21. (Quiz 1, Summer 2025) Consider a sample space S and three events A, B, C which partition the space into eight disjoint events, labelled D_1 to D_8, as shown in the following Venn diagram:



You are also given a probability measure $\mathbb{P}$ such that:

$$\mathbb{P}(A) = .4, \quad \mathbb{P}(A \cup B) = .6, \quad \mathbb{P}(A \cup B \cup C) = .7$$
$$\mathbb{P}(A \cap C^c) = .2, \quad \mathbb{P}(C \cap B) = .1$$

(a) (2 points) Use the rules of probability to find $\mathbb{P}(A \cap C)$.

(b) (3 points) Use the rules of probability to find $\mathbb{P}(A^c \cap B^c \cap C)$.

(c) (2 points) Find the *minimum* possible value for $\mathbb{P}(A \cap B)$, and provide the probabilities for each event $D_1, \ldots, D_8$ that result in that minimum.

(d) (3 points) Find the *maximum* possible value for $\mathbb{P}(A \cap B)$, and provide the probabilities for each event $D_1, \ldots, D_8$ that result in that maximum.

22. (Quiz 1 Summer 2025) A set of three dice, denote them A, B, and C, is called *intransitive* (or non-transitive) if it has the property that A is more likely to roll higher than B, and B is more likely to roll higher than C, but A is NOT more likely to roll higher than C. In other words, the set of dice is intransitive if the binary relation: "X is more likely to roll higher than Y" is not transitive on its elements. (reproduced from Wikipedia: Intransitive Dice)
Consider the following set of dice:

- Die A has sides $\{2, 2, 4, 4, 9, 9\}$.
- Die B has sides $\{1, 1, 6, 6, 8, 8\}$.
- Die C has sides $\{3, 3, 5, 5, 7, 7\}$.

You can readily verify that the probability of rolling die A higher than B is $5/9$, and similarly for $B > C$ and $C > A$ (see Wikipedia article above).
For this question, assume you *roll all three dice at once* and are interested in which one gives the *highest number overall*. Define the event H_A="die A is the highest overall", and similarly for H_B, H_C.

(a) (3 points) Are the events H_A, H_B, H_C a) mutually exclusive, or b) a partition? Justify your answer.

(b) (7 points) Find the probabilities $\mathbb{P}(H_A), \mathbb{P}(H_B), \mathbb{P}(H_C)$; are they all equal?